

# Kestrel 1.5T catalyst efficiency below threshold from degraded upstream oxygen

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Area: bulletins

## 1 Condition

Affected Kestrel 1.5T vehicles may set DTC P0420 (Catalyst System Efficiency Below Threshold, Bank 1) with the MIL illuminated. The catalytic converter assembly CAT-5500 is flagged by the monitor when downstream switching activity approaches upstream activity. Customers usually report no drivability complaint, only an emissions warning lamp. Converter construction and ratings are described in the Kestrel Emissions Service Manual :: Catalytic Converter (Bank 1 and Bank 2) .

## 2 Cause

In the majority of cases the CAT-5500 converter is functional and P0420 is set falsely because the upstream oxygen sensor, part number O2U-5100, has degraded. A lazy or slow-switching O2U-5100 distorts the fuel control signal the catalyst monitor relies on, causing an apparent efficiency loss. Replacing the converter without addressing the sensor will not resolve the fault. Sensor placement and signal characteristics are covered in the Kestrel Emissions Service Manual :: Upstream and Downstream Oxygen Sensors .

## 3 Diagnosis

Run procedure PROC-O2-SENSOR to evaluate the switching rate and response time of the O2U-5100 before condemning the converter. A sluggish upstream signal with normal downstream behavior confirms the sensor as the source of P0420. Compare both sensor traces against the expected waveform and review the catalyst monitor data in the Kestrel Emissions Service Manual :: Catalyst Efficiency Faults . General emissions diagnostic context is found in the Kestrel Emissions Service Manual :: Oxygen Sensing .

## 4 Repair Procedure

When the upstream sensor is confirmed degraded, replace the O2U-5100 per procedure PROC-O2-SENSOR and clear codes, then complete a catalyst monitor drive cycle to confirm P0420 does not return. Only if the fault persists with a verified-good sensor should the CAT-5500 converter be replaced following procedure PROC-CAT. Use the torque and anti-seize guidance in the Kestrel Emissions Service Manual :: Oxygen Sensor Replacement Procedure . Converter replacement steps are detailed in the Kestrel Emissions Service Manual :: Catalytic Converter Replacement Procedure .